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**Code No: 135SC**

**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD**
**B. Tech III Year I Semester Examinations, March - 2021**
**OPERATING SYSTEMS**
**(Common to CSE, IT)**

**Time: 3 hours** **Max. Marks: 75**

**Answer any five questions**
**All questions carry equal marks**
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1.a) Explain multiprocessing systems and write its advantages.
b) Write the services provided by Operating System to provide security and protection. [7+8]

2.a) Compare and contrast various structures of Operating System.
b) What are time sharing systems? Explain in detail. [8+7]

3.a) Explain Shortest Job First and FCFS CPU scheduling algorithms with examples.
b) What is monitor and Give a solution for Reader's and writer's Critical section problem using monitors. [7+8]

4.a) Explain about process control block with neat sketch.
b) Distinguish between ordinary pipes and named pipes in UNIX. [7+8]

5.a) Discuss about virtual memory and address translation in virtual memory.
b) Explain about LRU approximation and FIFO page replacement algorithms. [7+8]

6.a) Explain about Thrashing.
b) Explain various structures of page table. [8+7]

7.a) Discuss about SCAN and C-SCAN disk scheduling algorithms.
b) Give a note contiguous and indexed allocations method for files. [7+8]

8.a) Explain the methods for handling deadlock.
b) Explain about implementation of access matrix. [7+8]

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